Evaluate yourself : What is the major product obtained when 2,3 – dimethyl pentan -3 – ol is heated in the presence of H_2SO_4.

Oxidation of alcohols

The important reactions of alcohols are their oxidation to give carbonyl compounds. The commonly used oxidising agent is acidified sodiumdichromate. Oxidation of primary alcohols give an aldehyde which on further oxidation gives the carboxylic acids. To stop the oxidation reaction at the aldehyde / ketone stage, pyridinium chlorochromate (PCC) is used as an oxidising agent.

Example

$$\underset{\text{ethanol}}{CH_3-CH_2-OH} \xrightarrow[(O)]{\text{acidified } Na_2Cr_2O_7} \underset{\text{ethanal}}{CH_3-CHO} \xrightarrow[(O)]{\text{acidified } Na_2Cr_2O_7} \underset{\text{ethanoicacid}}{CH_3-COOH}$$

$$\underset{\text{Propan - 2- ol}}{CH_3-\underset{\displaystyle OH}{\underset{|}{CH}}-CH_3} \xrightarrow[(O)]{\text{acidified } Na_2Cr_2O_7} \underset{\text{Propanone}}{CH_3-\overset{\displaystyle O}{\overset{\|}{C}}-CH_3} \xrightarrow[(O)]{\text{acidified } Na_2Cr_2O_7} \underset{\text{ethanoicacid}}{CH_3-\overset{\displaystyle O}{\overset{\|}{C}}-OH}$$

$$\underset{\text{Propan - 1- ol}}{CH_3-CH_2-CH_2-OH} \xrightarrow{PCC} \underset{\text{Propanal}}{CH_3-CH_2-CHO}$$

Tertiary alcohols do not undergo oxidation reaction under normal conditions, but at elevated temperatures, under strong oxidising agent cleavage of C –C bond takes place to give a mixture of carboxylic acid.

Swern oxidation

In this method, dimethyl sulfoxide (DMSO) is used as the oxidising agent, which converts alcohols to ketones / aldehydes.

In this method an alcohol is treated with DMSO and oxalyl chloride followed by the addition of triethylamine.

$$\underset{\text{Propan - 2- ol}}{CH_3-\underset{\displaystyle OH}{\underset{|}{CH}}-CH_3} + \underset{\text{DMSO}}{H_3C-\overset{\displaystyle O}{\overset{\|}{S}}-CH_3} + \underset{\text{Oxalylchloride}}{Cl-\overset{\displaystyle O}{\overset{\|}{C}}-\overset{\displaystyle O}{\overset{\|}{C}}-Cl} \xrightarrow{Et_3N} \underset{\text{propanone}}{CH_3-\overset{\displaystyle O}{\overset{\|}{C}}-CH_3} + (CH_3)_2S + CO_2 + CO + 2HCl$$

XII U11-Hydroxy compounds.indd 117 16-12-2022 19:42:04